

HR Analytics Dashboard

Power BI Project Report & Summary

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1. Executive Summary

This project analyzes an HR dataset of 183 employees across India and New Zealand using Power BI. A single-page interactive dashboard was built with four visuals covering headcount, gender split, age distribution and salary by department, plus a Country slicer to compare the two countries.

- Procurement (55) and Website (54) are the largest departments; HR is the smallest (8).
- The 30-34 age group is the biggest (67 employees); most staff are between 25 and 39.
- HR has the highest average salary (\$89,650) and Sales the lowest (\$67,866).
- Company-wide average salary is \$77,174, ranging from \$33,920 to \$119,110.

2. Dataset

Item	Detail
Source file	hr-data.xlsx (sheet: Data)
Rows / columns	183 / 8
Columns	Name, Gender, Age, Rating, Date Joined, Department, Salary, Country
Departments	Procurement, Website, Finance, Sales, HR
Countries	India (IND) 92, New Zealand (NZ) 91
Data quality	No missing values, no duplicate names

3. Approach

- Imported the Excel file into Power BI Desktop.
- Created DAX measures: Head count, Min Salary, Max Salary, Average Salary.
- Built a custom Age bin group (bin size 5) for the age histogram.
- Applied conditional formatting (rules) to highlight larger age groups.
- Added a Country slicer, a report title and a dark theme for a consistent look.

4. Visuals and Findings

Visual 1: Headcount by Department. Column chart, Department on the X-axis and the Head count measure on the Y-axis.

Department	Headcount
Procurement	55
Website	54
Finance	38
Sales	28
HR	8

Visual 2: Headcount by Department and Gender. Clustered column chart with Gender as legend. The dataset has 6 employees in the "Other" category, which were filtered out so the chart compares Male and Female only.

Department	Female	Male
Procurement	23	32
Website	27	24
Finance	19	16
Sales	11	17
HR	5	3

Visual 3: Age Distribution (histogram). Stacked column chart using an Age bin group of size 5. Bars for age groups with 20 or more employees are highlighted through conditional formatting.

Age group	Employees
15-19	2
20-24	30
25-29	44
30-34	67
35-39	24
40-44	14
45-49	2

Visual 4: Salary Summary by Department. Table with Min, Max and Average Salary measures (0 decimals) and a Country slicer to view each country separately.

Department	Min	Max	Average
Procurement	\$37,920	\$115,920	\$81,794
Website	\$33,920	\$118,840	\$78,754
Finance	\$48,170	\$115,440	\$72,473
Sales	\$36,040	\$112,780	\$67,866
HR	\$45,510	\$119,110	\$89,650
Total	\$33,920	\$119,110	\$77,174

5. Key Insights

- Headcount is concentrated: Procurement and Website together make up about 60% of employees.
- Gender balance varies by department: Sales and Procurement lean male, while Website and HR are closer to even or lean female.
- The workforce is young: most employees are aged 25-39.
- Pay differs noticeably across departments, but HR has only 8 people, so its high average should be read with caution.

6. Limitations

- Salary currency is not specified in the source data; values are shown with a \$ sign only for formatting.
- India and NZ show identical salary ranges, which suggests the dataset is synthetic, so country comparisons are illustrative.
- Small groups (HR, the "Other" gender category) limit how strong the conclusions can be.
- The dashboard covers four analyses; performance ratings and the hiring trend are not yet included.

7. Next Steps

- Add KPI cards (total employees, average salary, average age).
- Add the remaining analyses: performance rating spread, hiring trend over time, top earners per country, and rating vs. salary.
- Publish the report, add a screenshot to the GitHub README, and build a data pipeline project (Python/SQL) to feed a similar dashboard.

8. Tools Used

Power BI Desktop, DAX, Excel.